SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: A Centralized Platform for Educational and Career Opportunities

GROUP NUMBER and MEMBERS: Group No: 5 / Members: Kaan DÖNMEZ, Hilal Seray ORTAN, Baran Deniz AYDIN, Salih Batuhan ŞENER, Doruk YAŞAR, Selin TAY, Ömer ŞEN

Questions to identify measurements:
• **Did each sprint finish according to the planned schedule?** • **How much effort did the team spend on each feature and sprint?** • **How many tasks were completed in each sprint?** • **How many defects were found during testing?** • **How effective were the test cases for the main features?** • **Did the system performance improve after optimization?** • **How many requirement or feature changes occurred during the project?** • **Were important quality areas such as security and response time completed before the final sprint ended?**
Identified measurements:
• **Planned sprint / actual completion sprint for each feature** • **Planned task completion date / actual task completion date** • **Number of completed tasks per sprint** • **Weekly development hours and weekly testing hours** • **Number of defects found and number of critical defects** • **Total test cases / passed test cases / failed test cases** • **Test pass rate = passed test cases / total test cases x 100** • **Number of requirement or feature changes** • **Average system response time before and after optimization** • **Number of open and closed issues in the project tracking tool**
Measurement storage and collection:
• **What: Planned and actual completion information for Sprint 1, Sprint 2, Sprint 3, and Sprint 4** **When: At the end of each sprint** **Format: Sprint name, planned date, actual date, completed/not completed** **How: Recorded in the project schedule or shared spreadsheet by the project manager** • **What: Development and testing hours** **When: At the end of each week** **Format: Integer or real number data** **How: Each team member records hours in a shared spreadsheet** • **What: Defects, critical defects, and change requests** **When: Whenever a problem or change is found** **Format: Integer counts and short descriptions** **How: Recorded in GitHub Issues, Trello, or a defect/change log** • **What: Test case results and response time** **When: After each testing session and after optimization work** **Format: Total, passed, failed test cases and response time values** **How: Recorded by the testing member in the testing spreadsheet**

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Measurement Type	Description	Example Measurements
Schedule / Sprint Progress	Compares planned sprint dates with actual completion dates and measures how many planned tasks are completed in each sprint.	User Registration and Authentication was planned to be completed between 25.02 and 01.03, and the actual completion date will be recorded to measure schedule accuracy. In the second sprint, 5 out of 6 planned tasks including Search and Filter, Personalized Feed, and Application Tracking were completed, resulting in an 83% completion rate.
Effort	Measures planned versus actual time spent on development, testing, meetings, documentation, and bug fixes. Tracks effort distribution across activity types and compares estimation accuracy across sprints to evaluate whether the team is improving its planning over time.	The AI-Based Recommendation Engine was estimated at 20 person-hours but actually required 28 person-hours, resulting in a 40% estimation variance. The actual effort broke down as 18 hours of development, 5 hours of testing, 3 hours of bug fixing, and 2 hours of sprint planning meetings. Compared to the first sprint where Database Design exceeded its estimate by 55%, the estimation accuracy improved noticeably by the third sprint, indicating that the team became more realistic in their planning as the project progressed.
Quality / Defects	Measures product quality by tracking the number and severity of defects found during development and testing phases.	During the development of the Notification System, 4 critical defects related to push notification delays were discovered. The Deadline Reminder module had 2 medium-severity defects caused by incorrect timezone calculations, and Duplicate Prevention had 1 minor UI label mismatch.
Test Pass Rate	Shows the percentage of successful test cases out of total test cases executed for each feature.	The Search and Filter Functionality passed 42 out of 48 test cases, achieving an 87.5% pass rate. User Registration and Authentication performed better with 30 out of 32 test cases passing, reaching a 93.75% success rate.
Change Tracking	Tracks the number of requirement modifications, feature additions, or scope changes requested during the project lifecycle.	Three scope changes were introduced during the second sprint based on instructor feedback: the Language Level Filter was added as a new feature, Application Tracking was raised from Medium to High priority, and the Career Success Stories module was added to the project objectives.
Performance	Measures system response time and loading speed before and after optimization efforts.	The Personalized Feed initially had an average response time of 4.2 seconds. After implementing database indexing and query caching during the fourth sprint, the response time improved to 1.8 seconds.
Code Coverage	Measures the percentage of source code that is executed by automated tests, indicating how thoroughly the codebase is tested.	The NestJS backend unit tests for the Search and Filter module achieved 78% code coverage after the second sprint. The target is to reach 85% overall backend coverage by the end of the fourth sprint through additional integration tests.
Risk Exposure	Tracks identified project risks, their likelihood, impact, and resolution status throughout the project.	Five risks were identified at the beginning of the project. The API rate limiting issue was classified as high likelihood and high impact, and was mitigated by implementing a caching layer. PostgreSQL connection pooling was resolved by configuring PgBouncer, while Flutter version compatibility remained under monitoring as a low-likelihood risk.
User Satisfaction	Measures end-user satisfaction through usability testing scores and feedback collected during demo and review sessions.	A usability test conducted with 10 pilot users at the end of the fourth sprint resulted in a System Usability Scale score of 74 out of 100. Users rated Search and Filter 4.2 out of 5, AI Recommendation accuracy 3.8 out of 5, and overall navigation 4.5 out of 5.